

User Story: As an admin, I want to see popular item categories so that I understand trends.

UAT 1 — View Popular Categories as Admin

- Given I am logged in as an Admin
- And completed transactions exist in multiple categories
- When I navigate to the admin dashboard
- Then I should see a horizontal bar chart titled "Popular Categories"
- And categories should be sorted by transaction count descending (most popular first)

UAT 2 — View Category Distribution Pie Chart

- Given I am logged in as an Admin
- And completed transactions exist in multiple categories
- When I view the "Category Distribution" chart
- Then I should see a pie chart showing the top 5 categories by total value
- And remaining categories should be grouped as "Other"

UAT 3 — See Zero-Transaction Categories

- Given I am logged in as an Admin
- And no transactions exist in "Stationary" or "Tickets"
- When I view the popular categories chart
- Then "Stationary" and "Tickets" should still appear with a count of 0

UAT 4 — View Tooltip with Count and Value

- Given I am logged in as an Admin
- And the categories chart is displayed
- When I hover over a category bar
- Then I should see a tooltip showing the category name, transaction count, and total value

UAT 5 — Filter Transaction Trends by Period

- Given I am logged in as an Admin
- And the trends chart is displayed
- When I click "Month"
- Then the line chart should re-render grouped by month
- And switching back to "Week" should re-render grouped by week

UAT 6 — View Summary Statistics

- Given I am logged in as an Admin
- When I view the top of the dashboard
- Then I should see stat cards for Total Transactions, Total Value, and Avg Value/Day

UAT 7 — Export Categories as CSV

- Given I am logged in as an Admin
- And the dashboard has loaded
- When I click "Export CSV"
- Then a categories.csv file should download with columns Category, Transaction Count, Total Value
- And a transactions.csv file should also download

UAT 8 — Non-Admin Cannot Access Analytics

- Given I am logged in as a Student
- When I navigate to /analytics/categories
- Then I should receive a 403 response
- And I should not see any analytics data

UAT 9 — Unauthenticated User Cannot Access Analytics

- Given I am not logged in
- When I navigate to /analytics/categories
- Then I should receive a 403 response

UAT 10 — Dashboard Loads with No Data

- Given I am logged in as an Admin
- And there are no completed transactions in the system
- When I navigate to the admin dashboard
- Then all categories should display with a count of 0
- And summary stats should show 0
- And the trends chart should be empty

User Story: As a user, I want to rate and leave a review for another user after a transaction so that I can share my experience and help build trust within the platform

UAT 1 — Open rating page

- Given I click the Rate button from Transaction History
- When the rating page opens
- Then I should see the seller rating interface displayed correctly

UAT 2 — Display seller information

- Given a valid sellerId is passed through the URL
- When the rating page loads
- Then the correct seller information should be displayed

UAT 3 — Select a star rating

- Given the rating stars are displayed
- When I click on a star
- Then the selected star and all previous stars should highlight visually

UAT 4 — Submit a review

- Given I selected a rating and entered a review message
- When I click the submit button
- Then the rating and review should be submitted successfully

UAT 5 — Validate empty rating submission

- Given I have not selected a rating
- When I click the submit button
- Then the review should not be submitted
- And I should see a validation message

UAT 6 — Display success feedback

- Given the review submission is successful
- When the backend confirms the rating was saved
- Then I should see a success message displayed on the page

UAT 7 — Responsive ratings layout

- Given I open the rating page on different screen sizes
- When the page loads
- Then the rating card, stars, and form elements should remain properly aligned and readable

User Story: As a student buyer, I want to purchase or reserve an item so that i can initiate a transaction with the seller

UAT 1 — Create listing

- Given I am authenticated
- When I submit valid listing details and a valid image
- Then a new listing should be created
- And the response should return the created listing

UAT 2 — Reject unauthenticated listing creation

- Given I am not authenticated
- When I try to create a listing
- Then the request should be rejected

UAT 3 — Validate missing fields on creation

- Given I am authenticated
- When I submit a listing without title, description, price, or image
- Then the listing should not be created
- And an error should be returned

UAT 4 — View all listings

- Given listings exist
- When I request /listings/all
- Then I should receive all listings ordered by newest first

UAT 5 — View latest listings

- Given listings exist
- When I request /listings
- Then I should receive the latest 6 listings

UAT 6 — Search listings

- Given listings exist
- When I search using keyword/category/condition/price filters
- Then only matching listings should be returned

UAT 7 — View one listing

- Given a listing exists
- When I request /listings/:id
- Then the correct listing should be returned

UAT 8 — Edit listing

- Given I am authenticated and a listing exists
- When I send valid updated details
- Then the listing should be updated

UAT 9 — Reject invalid update

- Given I am authenticated
- When I update a listing with an invalid category/condition/sale type
- Then the update should be rejected

UAT 10 — Delete listing

- Given a listing exists
- When I delete it
- Then it should be removed successfully

UAT 11 — Reserve available listing sets reserved_by

- Given I am authenticated as a buyer
- And an available listing exists that I do not own
- When I reserve the listing
- Then the listing status should be updated to reserved
- And the listing reserved_by field should store my user ID
- And the response should confirm the reservation

UAT 12 — Prevent duplicate reservation

- Given I am authenticated as a buyer
- And the listing has already been reserved by another user
- When I try to reserve the same listing
- Then the request should be rejected
- And the listing reserved_by field should not be changed
- And no new reservation should be created

UAT 13 — Search listings by status

- Given listings exist with different statuses
- When I request /listings/search?status=available
- Then only listings with status available should be returned
- And reserved or unavailable listings should not be included

UAT 14 — Search listings by reserved buyer

- Given I am authenticated as a buyer
- And I have reserved one or more listings
- When I request /listings/search?status=reserved&reserved_by=myUserId
- Then only listings reserved by me should be returned
- And listings reserved by other users should not be included

UAT 15 — Search listings returns seller name and ratings

- Given listings exist with seller profile and rating information
- When I request listings from the search endpoint
- Then each listing should include the seller's name
- And each listing should include the seller's rating data
- And the frontend should be able to display the seller name and rating on the listing card

User Story: As a staff member, I want to track and update transaction progress so that marketplace exchanges can be properly managed and completed.

UAT 1: View all transactions

- Given I am logged in as Trade Facility Staff
- When I open the Transactions page
- Then I should see a list of transactions from the database
- And each transaction should show the listing, buyer name, seller name, transaction type, date, and status.

UAT 2: Filter transactions by status

- Given I am on the Transactions page
- When I click a status filter such as Pending, Waiting Drop-Off, Waiting Pick-Up, Completed

- Then only transactions matching that status should be displayed.

UAT 3: Search transactions

- Given I am on the Transactions page
- When I type a buyer name, seller name, or listing name into the search bar
- Then the transaction list should update to show only matching transactions.

UAT 4: View sale transaction progress

- Given a sale transaction exists
- When I view it on the Transactions page
- Then I should see the progress stages as Pending → Drop-Off → Pick-Up → Completed
- And completed stages should appear as filled/darker circles.

UAT 5: Confirm drop-off for sale

- Given a sale transaction is waiting for drop-off
- When I click Confirm Drop-Off
- Then the transaction should move to the Waiting Pick-Up stage
- And the progress indicator should update.

UAT 6: Complete sale transaction

- Given a sale transaction has already had drop-off confirmed
- When I click Mark Completed
- Then the transaction status should become Completed
- And it should appear under the Completed filter.

UAT 7: View trade transaction progress

- Given a trade transaction exists
- When I view it on the Transactions page
- Then I should see the trade progress stages as Pending → Exchange → Completed.

UAT 8: Confirm trade exchange

- Given a trade transaction is waiting for exchange
- When I click Confirm Exchange
- Then the transaction should be marked as Completed
- And the progress indicator should show all stages as completed.